GENX2 Porewater SO4/Cl

2025 Samples Run 1

Date QAQC'd: 2025-10-02

Contents

##Add Required Packages

0.1 Run Information

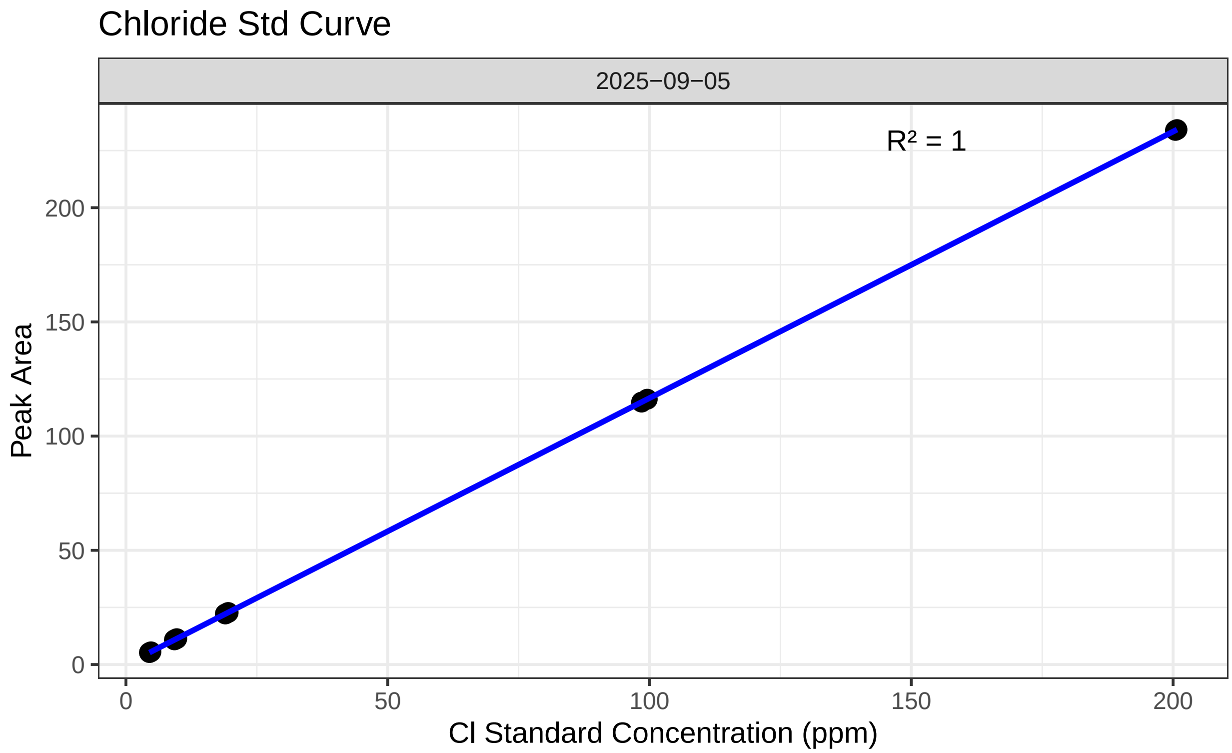
```
###### Run information - PLEASE CHANGE
  Date_Run = "2025-09-05"  #Date that instrument was run
  Year_run = "2025" #Year that instrument was run
  Run_by = "Zoe Read"  #Instrument user
  Script_run_by = "Zoe Read" #Code user
  run_notes = "Sample 7_GX2_50151 was above the SO4 detection limit
  (22 ppm vs the 20 ppm detection limit)
  " #any notes from the run

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
  raw_file_name_cl = "Raw Data/GenX2_Mesocosms2025_PW_Run1_20250905_Cl.txt"
  raw_file_name_so4 = "Raw Data/GenX2_Mesocosms2025_PW_Run1_20250905_SO4.txt"

#file path and name of processed data file
  processed_file_name = "Processed Data/GenX2_Mesocosms2025_PW_Processed_Cl_SO4_Run1_20250905.csv"

###### Log Files - PLEASE CHECK
#downloaded metadata csv of dilutions
  Raw_Metadata = "Raw Data/GenX2_Mesocosms2025_PW_Dilutions.csv"

#qaqc log file path for this year
  Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2024.csv"
```

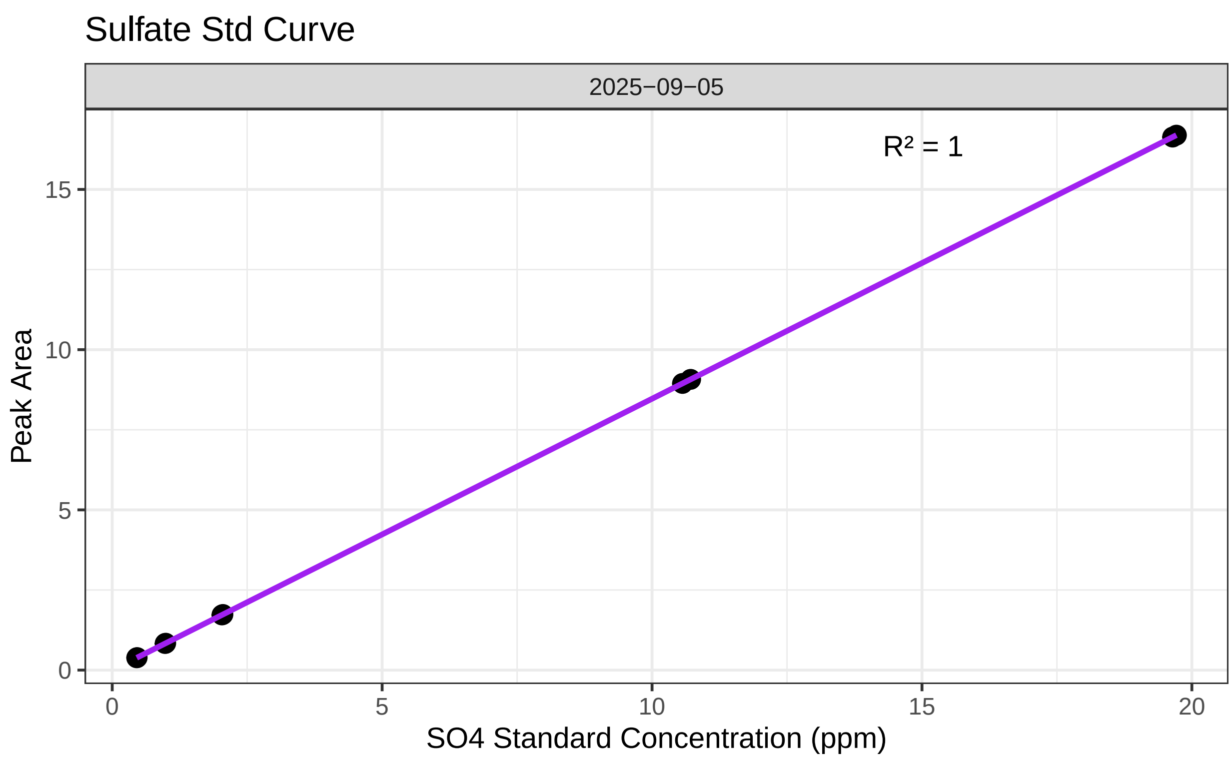
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

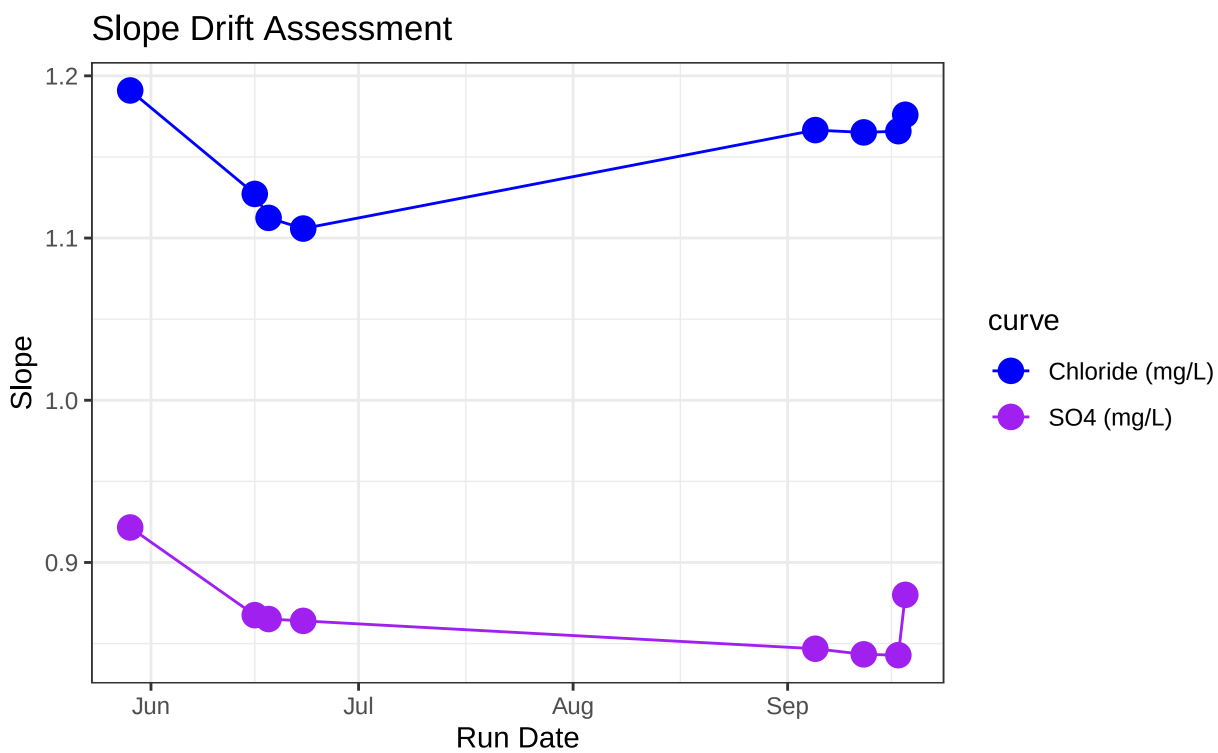
##Import Sample Data

0.2 Assess Standard Curves






```
## [1] "QAQC log file exists and has been read into the code."
```




```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

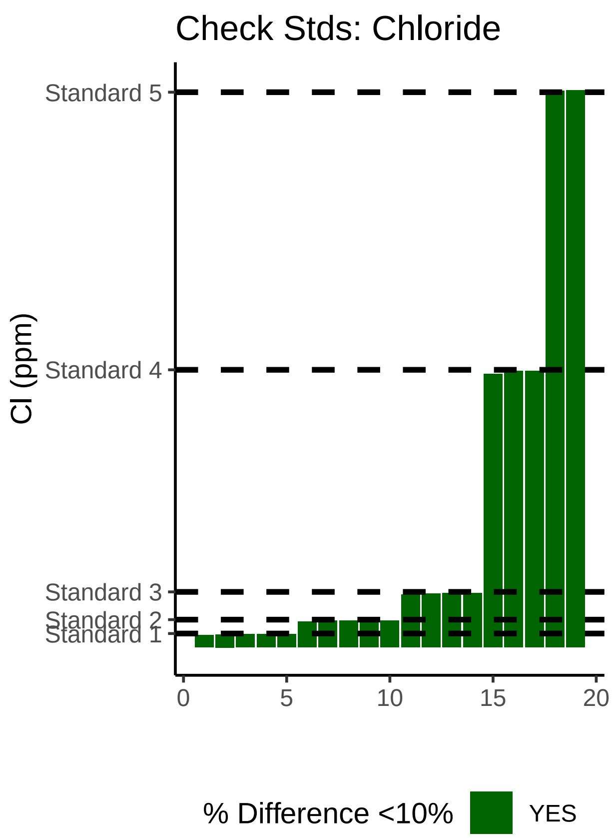
0.3 Assess Check Standards

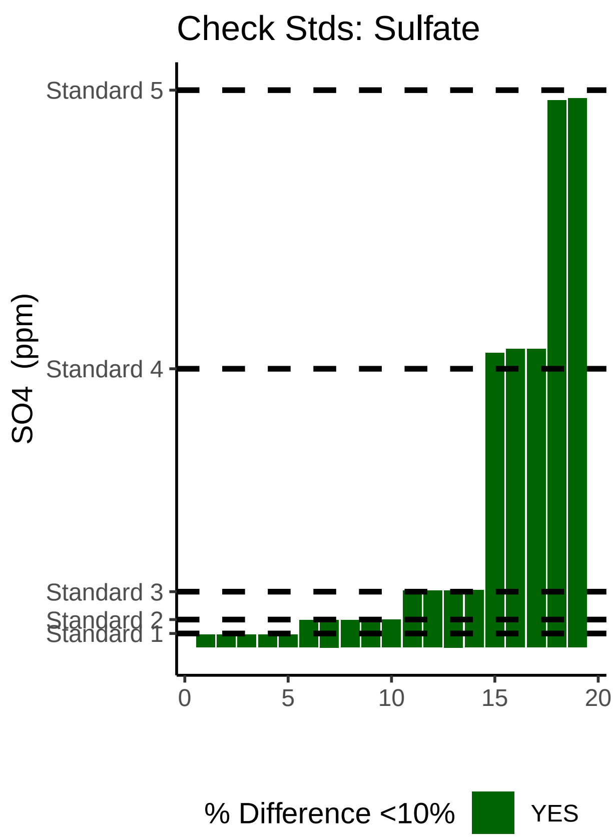
```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl sd_Cl   cv_Cl flag_Cl
##   <chr>        <dbl> <dbl>   <dbl> <chr>
## 1 Standard 1    4.69 0.115 0.0246  Chloride Check Standard RSD within Range - P~
## 2 Standard 2    9.62 0.200 0.0208  Chloride Check Standard RSD within Range - P~
## 3 Standard 3   19.4  0.265 0.0137  Chloride Check Standard RSD within Range - P~
## 4 Standard 4   99.2  0.622 0.00628 Chloride Check Standard RSD within Range - P~
## 5 Standard 5  201.   0.210 0.00104 Chloride Check Standard RSD within Range - P~
```

```
## # A tibble: 5 x 5
##   sample_ID  mean_SO4  sd_SO4  cv_SO4 flag_SO4
##   <chr>         <dbl>   <dbl>   <dbl> <chr>
## 1 Standard 1    0.459 0.00309 0.00673 Sulfate Check Standard RSD within Range --~
## 2 Standard 2    0.987 0.00421 0.00427 Sulfate Check Standard RSD within Range --~
## 3 Standard 3    2.04  0.00822 0.00403 Sulfate Check Standard RSD within Range --~
## 4 Standard 4   10.7   0.0868  0.00814 Sulfate Check Standard RSD within Range --~
## 5 Standard 5   19.7   0.0498  0.00253 Sulfate Check Standard RSD within Range --~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```





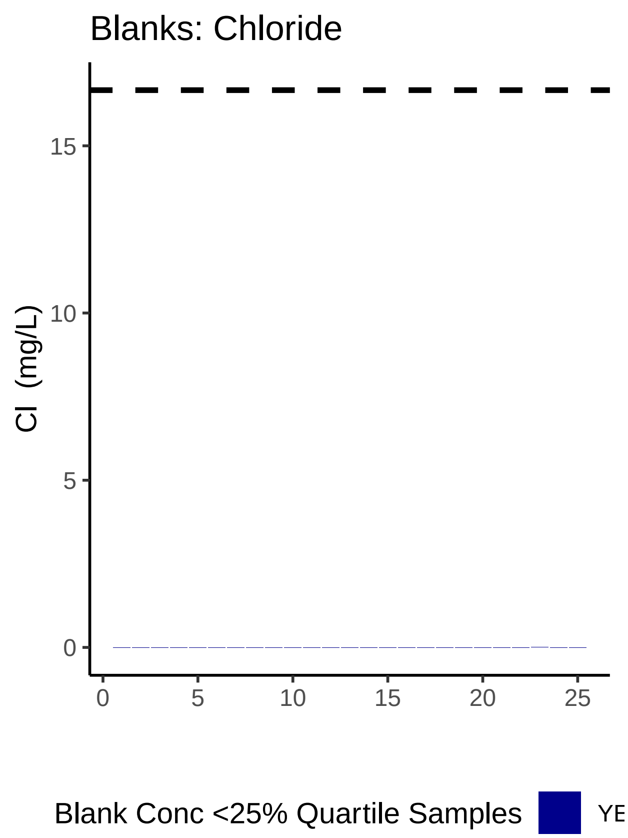

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

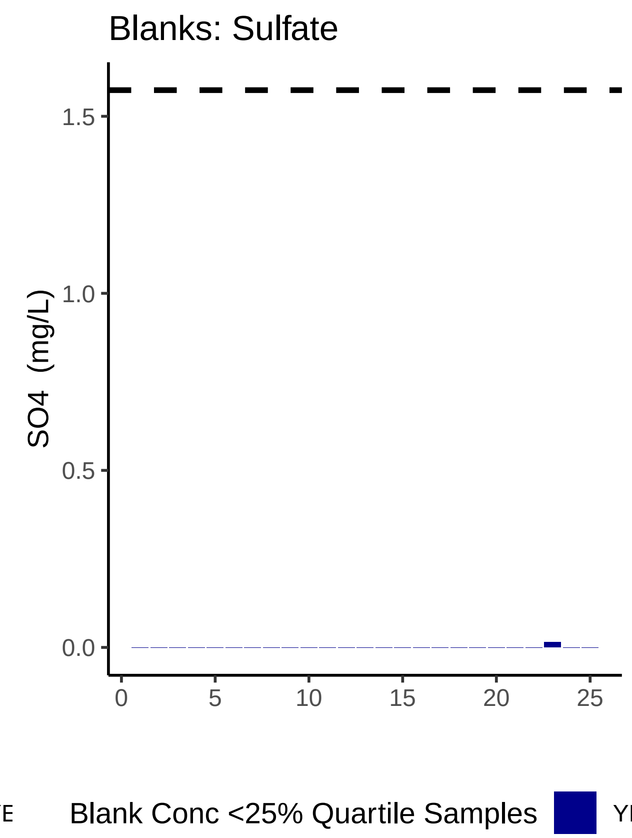
```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

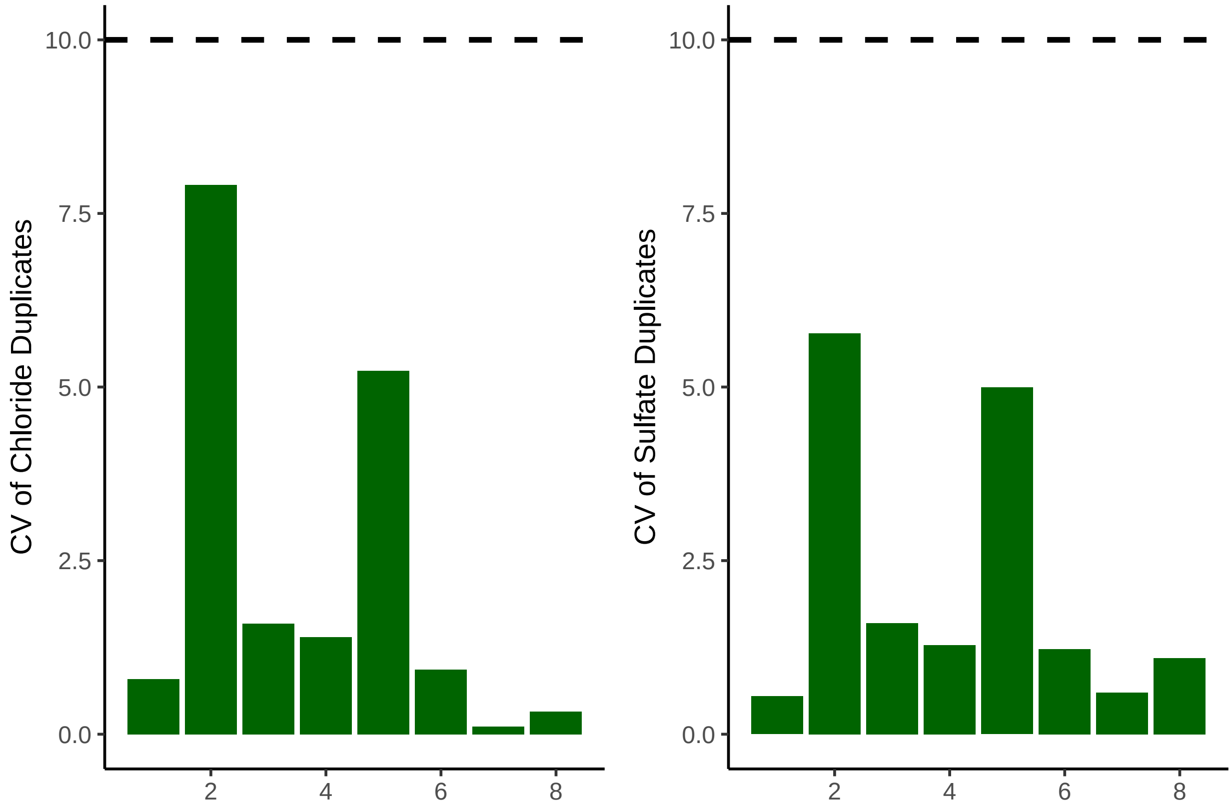
```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```





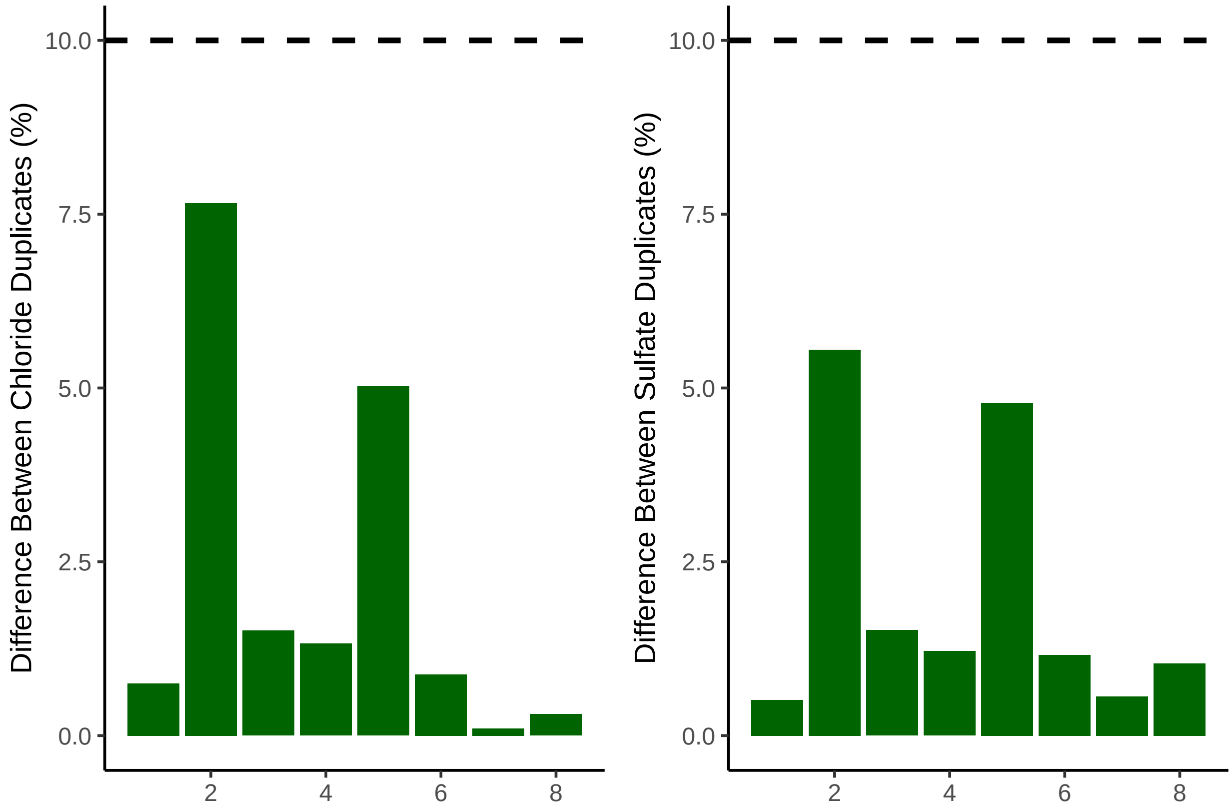

```
## Chloride blanks mean ppm:

## [1] 0.000428

## Sulfate blanks mean ppm:

## [1] 0.000592
```

0.5 Assess Duplicates




```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```




```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

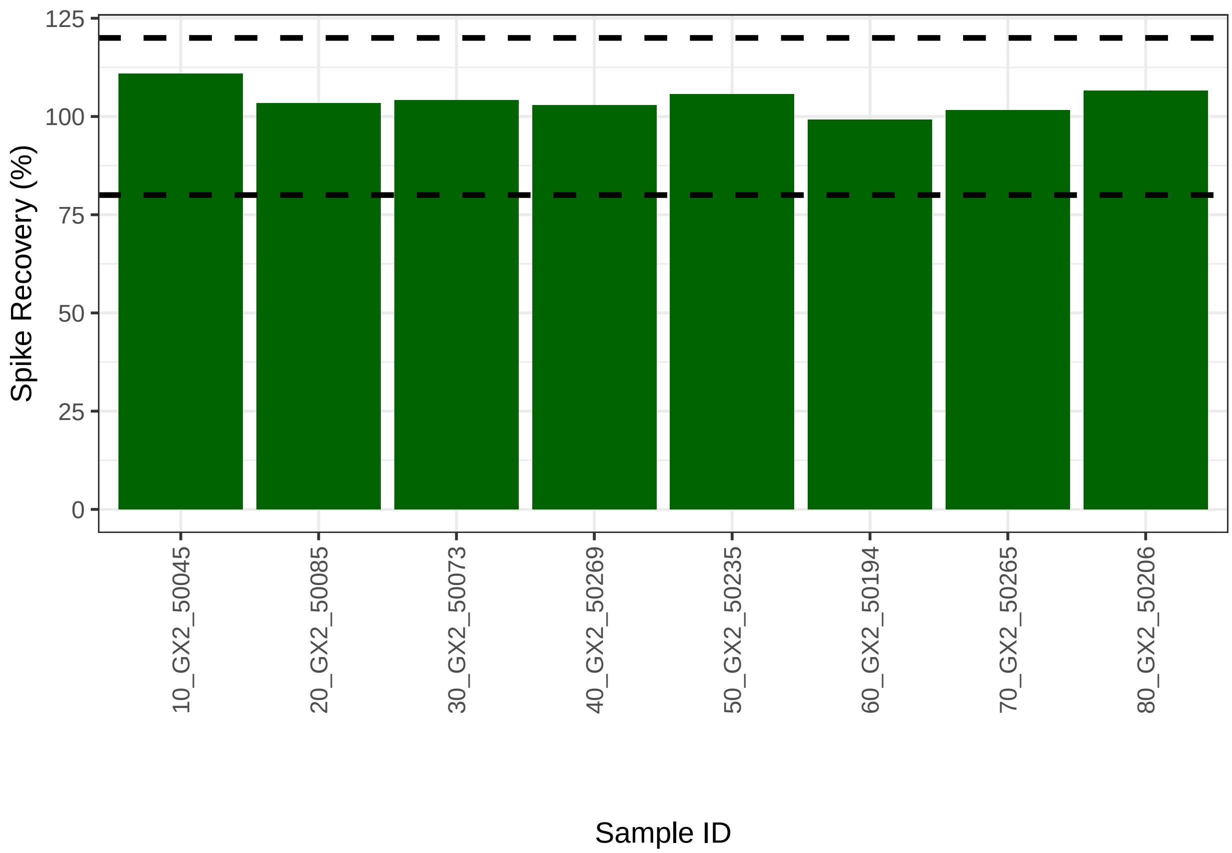
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```r
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples

#Separate namings
all_dat <- all_dat %>%
  mutate(sample_ID_long = sample_ID) %>%
  separate(sample_ID_long, into = c("number", "project", "ID", "test"), sep = "_") %>%
  mutate(sample_ID_short = paste(number, project, ID, sep = "_"))
```

```
## Warning: Expected 4 pieces. Missing pieces filled with `NA` in 124 rows [1, 2, 3, 4, 5,
## 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, ...].
```

```r
#merge dilution with sample run data
merged_data <- all_dat %>%
  left_join(dionex_metadata, by = c("sample_ID_short" = "Cl_SO4_ID"))
```

0.7 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

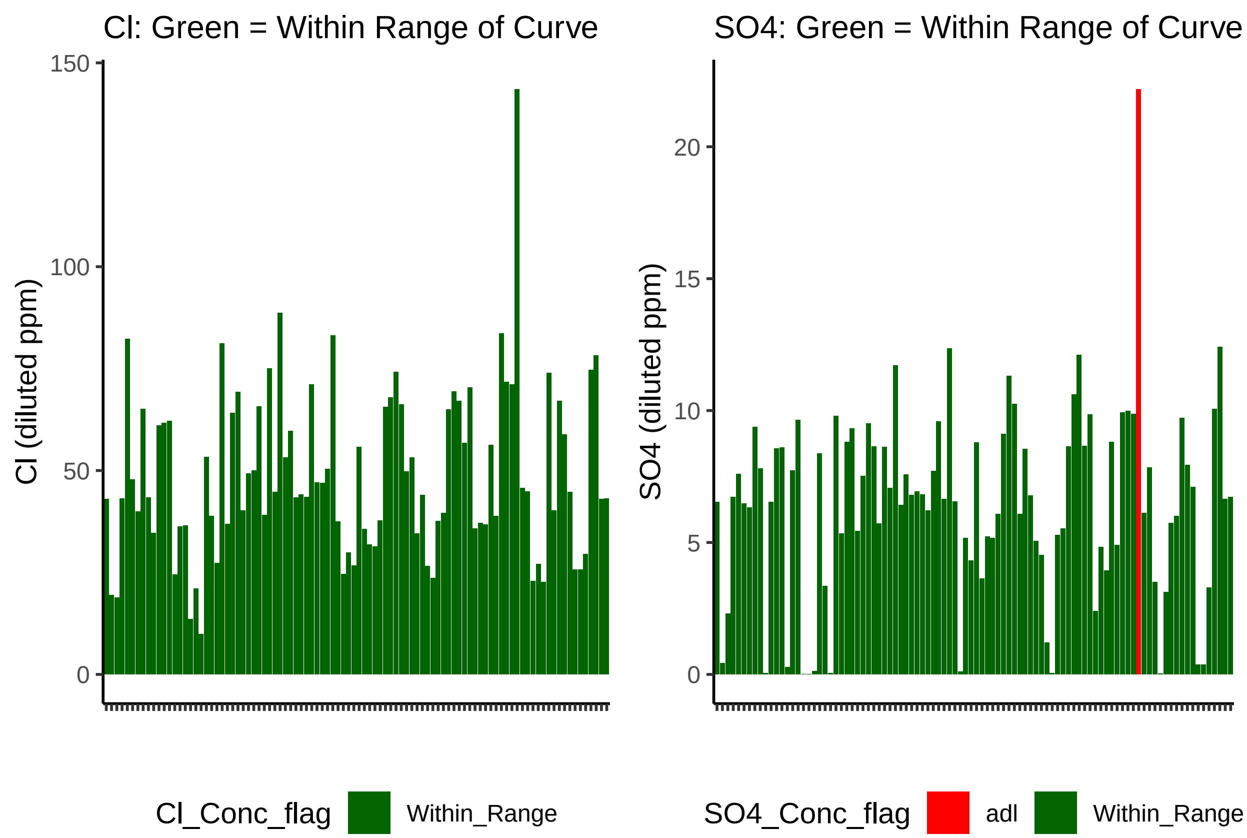


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	98.958333
adl	1.041667

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

0.9 Export Processed Data

#end